

Gauri Toshniwal

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EDUCATION

National Institute of Technology, Tiruchirappalli

Tamil Nadu, India

Masters of Technology - Data Analytics (CGPA: 9.58/10) [Rank 1]

08/2023 - 06/2025

Awarded Reliance Foundation Postgraduate Scholarship for the year 2023-24

Coursework: Deep Learning, NLP, Machine Learning, Federated Learning, Mathematics (Linear Algebra, Probability)

Government College of Engineering, Nagpur

Maharashtra, India

Bachelor of Engineering - Computer Science and Engineering (CGPA: 9.42/10)

06/2021

Coursework: Operating Systems, Data Structures & Algorithms, Artificial Intelligence, Databases, Discrete Mathematics, Compilers

RESEARCH

- **Forget Less, Solve More: Sequential Fine-Tuning with Adapter Shrinking for Math Word Problems** 07/2025
AI for Math 2025 @ ICML - (Accepted Poster), Master's Thesis (Paper)
 - Proposed a sequential fine-tuning approach with progressive adapter shrinking to reduce forgetting in LLMs (LLama, Qwen)
 - Built a full MATH-dataset pipeline with custom instruction formatting, label masking, LoRA (QLoRA 4-bit), DDP training, and an evaluation suite with greedy decoding, structured extraction, and EM scoring.
- **Contextual QA for Financial Causality Detection Combining Extractive and Generative Models** 12/2024
FinCausal 2025 @ COLING - 3rd Rank (English Track) (Paper)
 - Designed an extractive-generative hybrid system using RoBERTa-base for causal span detection and Gemma-2 9B for contextual justification generation.
 - Achieved **SAS 96.74%** and **EM 70.14%**, improving causal reasoning consistency in financial narratives.
- **Object Detection in Indian Food Platters using Transfer Learning with YOLOv4** 05/2022
DECOR 2022 @ ICDE - Best Student Paper Award (Paper)
 - Developed a domain-adapted YOLOv4 model and curated IndianFood10, a custom dataset for Indian cuisine recognition.

WORK EXPERIENCE

Astreya

Remote, India

AI/ML Engineer (Google Extended Workforce)

07/2025 - Current

- **CTI Verifier: High-Precision Ticket Classification & Preloading System**
 - Researched and developed a production-ready confidence-aware selective classifier with an abstention mechanism for **275-class** hierarchical ticket classification, systematically evaluating retrieval, embedding, and LLM-based approaches.
 - Built a confidence-calibrated classifier achieving **95.6% precision** on accepted predictions, enabling reliable automation of ~55K manual ticket classification decisions annually.
- **Corvus: Enterprise AI Copilot for IT Operations**
 - Built and productized an enterprise AI copilot, enabling SOP-grounded chat, next-step recommendations, quality audits, and context-aware AI workflows with Redis-backed state and BigQuery telemetry.
 - **Cut playbook search and ticket resolution time**, delivering **~10 FTE annual savings** through AI-powered automation, grounded recommendations, and quality-assurance workflows.

Findability Sciences

Remote, India

Data Scientist (Onsite - Hindustan Times Digital)

05/2021 - 08/2023

- **Demographics (Gender, Age, Interest) Prediction:**
 - Built user-level classifiers for gender, age, and interest prediction using grammatical and entity-based features from news headlines, achieving an F1 score of 0.65 for **2M** users daily. Eliminated reliance on third-party data, saving **₹500,000/month**.
- **Subscription Churn Prediction and Retention Feed:**
 - Developed XGBoost models (d-60, d-30, d-15) to track churn probability shifts and used **SHAP** for interpretability powering a personalized **retention feed** that improved user engagement.

CERTIFICATION

- **ACM Summer School on Responsible and Safe AI** [Certificate](#) 06/2024
- **AI Safety Fundamentals - Bluedot Impact** [Certificate](#) 03/2024 - 06/2024

SKILLS

Languages: Python, C, SQL

ML & AI: PyTorch, Scikit-learn, Pandas, NumPy, PySpark, Hugging Face Transformers, RAG, LLM Evaluation

Frameworks, Cloud & Tools: FastAPI, Flask, Git, Docker, GCP, Vertex AI, BigQuery ML, MongoDB, Redis

Areas of Interest: NLP, LLMs, Deep Learning, AI Evaluation, AI Safety